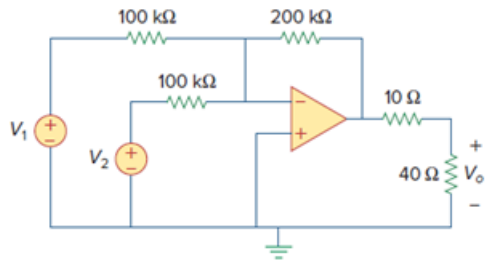


Problem

Referring to the circuit shown in Fig. 5.77, determine V_o in terms of V_1 and V_2 .

Figure 5.77:



Step-by-step solution

Step 1 of 4

Draw the circuit diagram as shown in Figure 1.

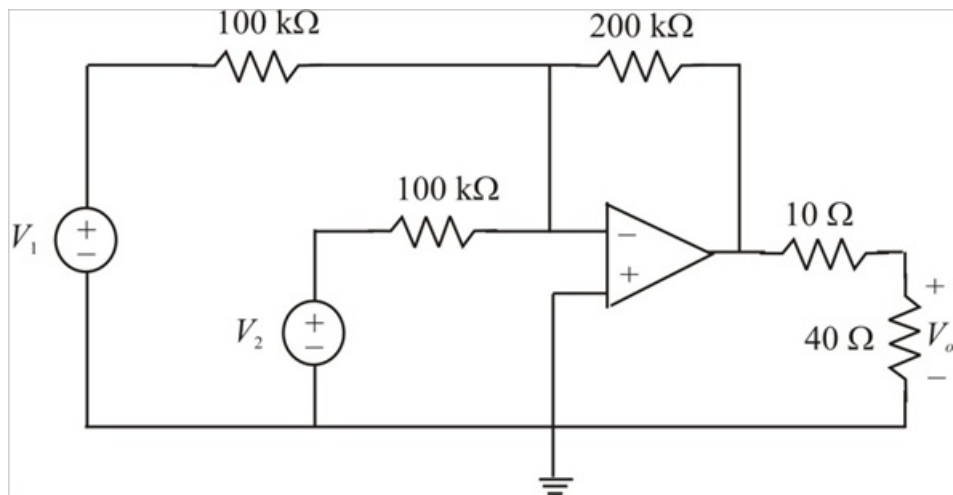


Figure 1

Step 2 of 4

The circuit is labeled as shown in Figure 2.

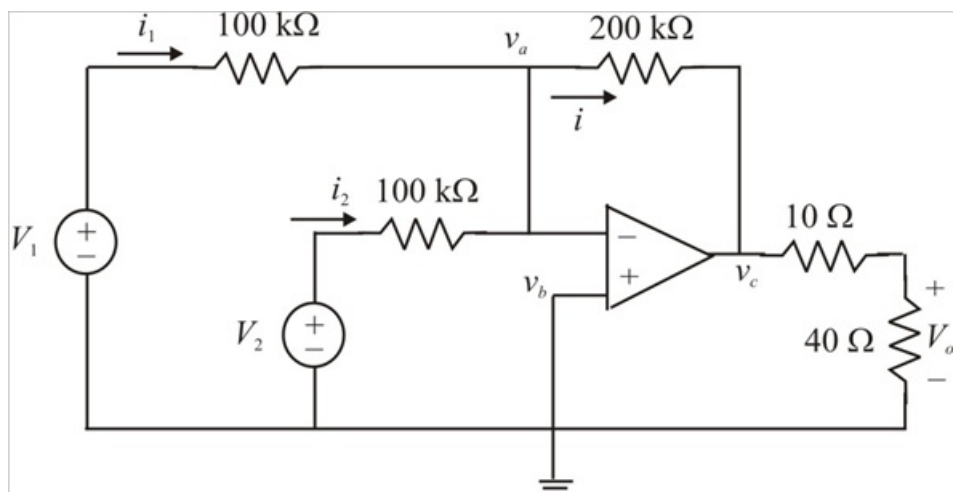


Figure 2

Step 3 of 4

Applying Kirchhoff's current law to node 1,

$$i = i_1 + i_2$$

$$\frac{v_a - v_c}{200} = \frac{V_1 - v_a}{100} + \frac{V_2 - v_a}{100}$$

$$v_a - v_c = 2V_1 - 2v_a + 2V_2 - 2v_a$$

$$5v_a - v_c = 2(V_1 + V_2)$$

$$\text{Therefore } 5v_a - v_c = 2(V_1 + V_2) \dots\dots (1)$$

But, $v_a = v_b = 0$ since non-inverting terminal is grounded.

Substituting $v_a = 0$ in equation (1), we get

$$5v_a - v_c = 2(V_1 + V_2)$$

$$5(0) - v_c = 2(V_1 + V_2)$$

$$v_c = -2(V_1 + V_2)$$

$$\text{Therefore } v_c = -2(V_1 + V_2) \dots\dots (2)$$

Step 4 of 4

Applying voltage division,

$$V_o = \frac{40}{40+10}(v_c)$$

$$= \frac{40}{40+10}(-2(V_1 + V_2))$$

$$= -1.6(V_1 + V_2) \text{ V}$$

Therefore the output voltage V_o in terms of V_1 and V_2 is $\boxed{-1.6(V_1 + V_2) \text{ V}}$.